

# BookStore MERN Project

## Documentation

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**Project Title:** BookStore – Online Book Shopping System

**Technology:** MongoDB, Express.js, React.js, Node.js (MERN)

### 1. Introduction

BookStore is a full-stack MERN web application that allows users to browse, search, purchase, review, and reserve books through a responsive interface. It includes separate user and admin modules with secure authentication.

### 2. Project Overview

Purpose: Develop an online bookstore with authentication, shopping cart, wishlist, checkout, reviews, reservations, and an admin dashboard.

Features:

- User Registration/Login
- Browse/Search Books
- Categories & Filters
- Wishlist
- Shopping Cart
- Checkout & Orders
- Reviews & Ratings
- Admin Dashboard
- Responsive UI

### 3. Architecture

Frontend: React.js, React Router, Axios, Bootstrap.

Backend: Node.js + Express.js REST APIs.

Database: MongoDB collections for Users, Books, Categories, Orders, Reviews, Wishlist, Cart, Reservations and Announcements.

## 4. Setup Instructions

Prerequisites:

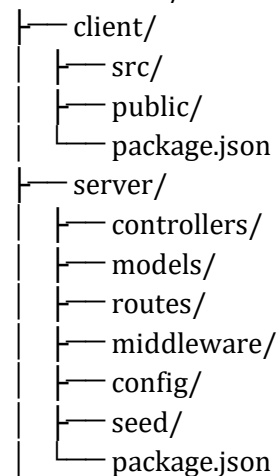
- Node.js
- MongoDB Community Server / MongoDB Atlas
- VS Code

Installation:

1. Clone the repository.
2. `cd server && npm install`
3. `cd client && npm install`
4. Configure `.env`.
5. Start MongoDB.
6. Run backend and frontend.

## 5. Folder Structure

BOOKSTORE/



## 6. Running the Application

Backend:

```
cd server
npm run dev
```

Frontend:

```
cd client
npm start
```

Seed Database:

```
cd server
node seed/seed.js
```

## 7. API Documentation

Authentication:

POST /api/auth/register

POST /api/auth/login

Books:

GET /api/books

GET /api/books/:id

Cart:

GET /api/cart

POST /api/cart

Wishlist:

GET /api/wishlist

POST /api/wishlist

Orders:

POST /api/orders

GET /api/orders

Admin:

GET /api/admin/users

GET /api/admin/books

PUT /api/admin/books/:id

DELETE /api/admin/books/:id

## 8. Authentication

JWT authentication is used. Passwords are hashed using bcryptjs. Protected routes require a valid JWT. Admin-only routes are protected using role-based authorization.

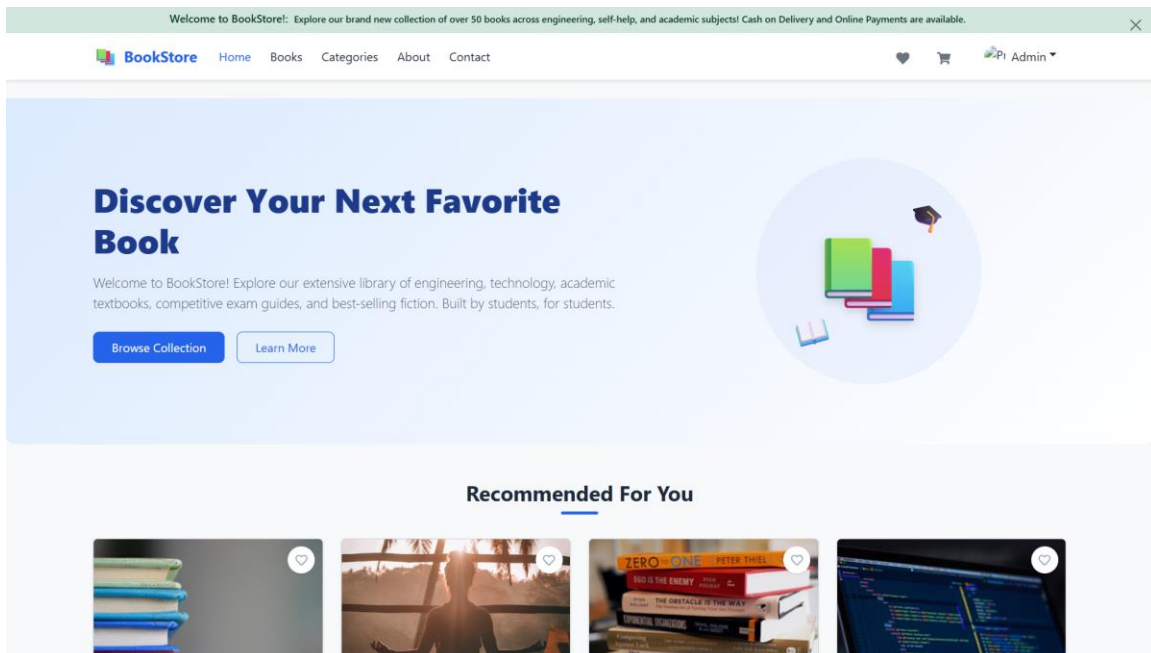
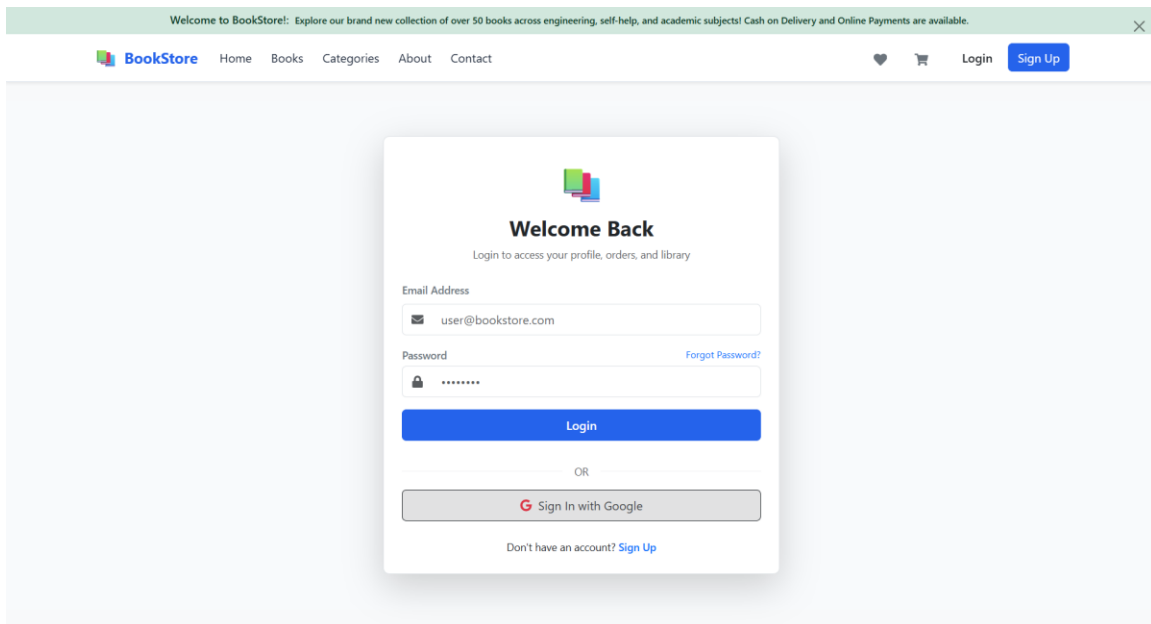
## 9. User Interface

Main pages include Home, Browse Books, Categories, Book Details, Wishlist, Cart, Checkout, Orders, Profile, Login, Signup, and Admin Dashboard. Insert screenshots here before submission.

## 10. Testing

Manual testing was performed for login, CRUD operations, book browsing, cart, wishlist, orders, and admin features. MongoDB was used to validate data persistence.

## 11. Screenshots / Demo



Welcome to BookStore! Explore our brand new collection of over 50 books across engineering, self-help, and academic subjects! Cash on Delivery and Online Payments are available.

BookStore

HomeBooksCategoriesAboutContact

Pi Admin

Our Book Collection

Search by title, author, or category...

Search

Sort By: Newest Arrivals

Self-Help

ebook

The Subtle Art of Not Giving a F\*ck

By Mark Manson

★ 0.0 (0)

₹399 ₹499 (20% OFF)

Add to Cart

Self-Help

Atomic Habits: An Easy & Proven Way to Build Good Habits &...

By James Clear

★ 5 (1)

₹499 ₹650 (23% OFF)

Add to Cart

Self-Help

How to Win Friends and Influence People

By Dale Carnegie

★ 0.0 (0)

₹249 ₹350 (29% OFF)

Add to Cart

Cooking

Salt, Fat, Acid, Heat: Mastering the Elements of Good Cooking

By Samin Nosrat

★ 0.0 (0)

₹799 ₹999 (20% OFF)

Add to Cart

BookStore

HomeBooksCategoriesAboutContact

Pi Admin

Book Categories

Filter categories by name...

Academic

Textbooks and educational literature.

Artificial Intelligence

Neural networks, machine learning, and future technology.

Biography

Life stories of notable individuals.

Business

Economics, finance, and entrepreneurship.

Children's Books

Bedtime stories and early learning.

Comics

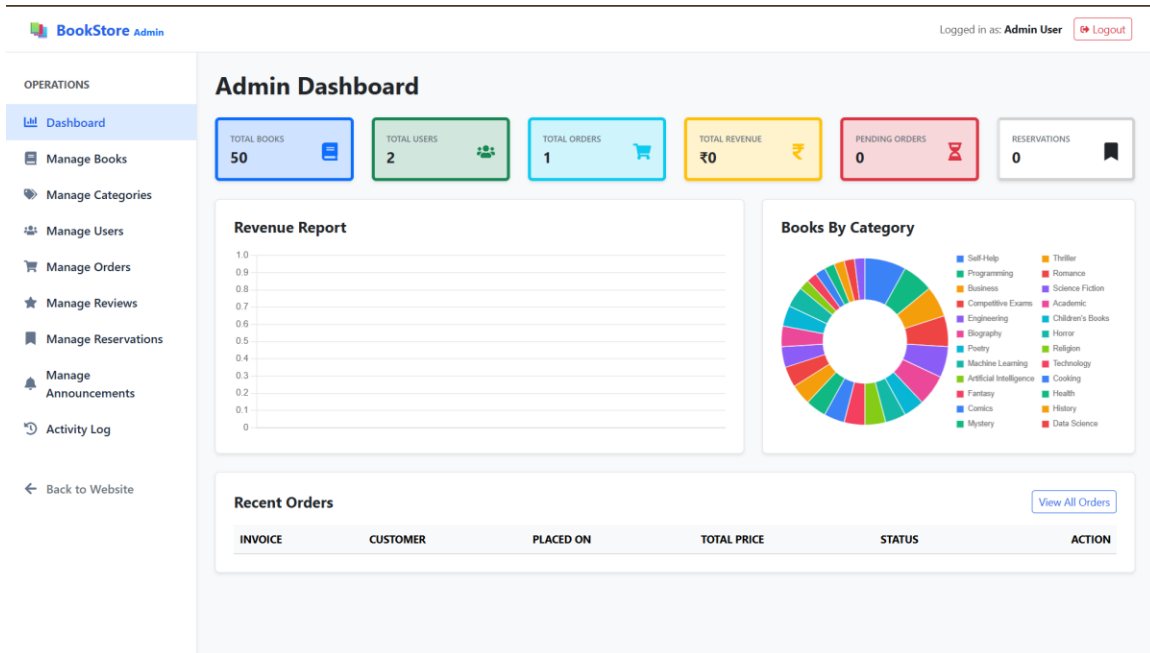
Graphic novels and illustrated stories.

Competitive Exams

Study guides for GATE, JEE, UPSC, etc.

Cooking

Recipes, culinary skills, and cuisines.



## 12. Known Issues

- Placeholder/fake book image URLs were replaced or need replacement.
- Minor runtime issues may require additional fixes.

## 13. Future Enhancements

- Online payment improvements
- Email notifications
- AI-based recommendations
- Inventory analytics
- Dark mode
- Multi-language support
- Real-time order tracking